PRACHI ZALA

Software & Applied AI Developer | Python & Data Analytics Specialist

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PROFESSIONAL SUMMARY

High-performing BCA developer (CGPA 9.12) and 6× Hackathon Team Leader specializing in test-driven Python backends, GraphRAG generative AI systems, computer vision, and quantitative data analytics. Proven track record of captaining cross-functional teams to 2nd Place in the SIH 2026 Internal Hackathon and leading squads across national and global arenas (HACKCULT '26, Hackbriven, SOU). Focused on production-grade engineering with 35+ automated Pytest test suites, Docker Compose, and data-backed architectural decision making.

EDUCATION

Shanti Business School, Ahmedabad (GTU) | *Bachelor of Computer Applications (BCA)* **2024 – 2027**

CGPA: 9.12 / 10.0 | Coursework: Data Structures & Algorithms, DBMS (SQL), Python Programming, Software Engineering, Networks.

Satyam Vidhyalaya, Ahmedabad | *Higher Secondary (HSC) & Secondary (SSC)* **2020 – 2024**

12th Grade (HSC): 71% (2024) • 10th Grade (SSC): 70% (2022)

CORE TECHNICAL SKILLS

Languages: Python 3.11, SQL, TypeScript, JavaScript (ES6+), Java 8+, HTML5, CSS3

Backend & Systems: FastAPI, Flask, Node.js, Express, SQLAlchemy, RESTful APIs, Docker Compose, Pytest (35 Tests), Redis RQ, JWT, Argon2

AI, RAG & Data Science: GraphRAG, Neo4j (Cypher), PostgreSQL (pgvector), ChromaDB, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Jupyter, XGBoost, SHAP, Ollama

Mobile & Frontend: React Native (Expo), Next.js 14/15, React, Tailwind CSS, Android SDK (Java), PWA, Firebase (RTDB & Firestore)

Developer Tools: Git, GitHub Actions, Postman, Linux/Bash, VS Code, Test-Driven Development (TDD)

HACKATHON & LEADERSHIP TRACK RECORD (6× TEAM LEADER)

SIH 2026 Internal Hackathon | *Team Leader — Team CyberForge* **2nd Place (58 Marks) • 2026**

- Captained a 6-member engineering squad to 2nd position in the university-wide SIH selection round.

HACKCULT '26 Hackathon (IIT Gandhinagar × NxtWave) | *Team Leader — Team Believers* **March 2026**

- Led Team Believers to build Dharohar 4D, an interactive 3D monument digital twin platform with Three.js and WebXR AR.

Hackbriven National-Level Hackathon (Ahmedabad) | *Team Leader* **2026**

- Spearheaded technical sprint for Qoneqt-ContentOS, an automated content factory and publishing orchestration platform.

Silver Oak University (SOU) Mini Hackathon | *Team Leader* **2025**

- Captained team to build MathSprint, an anti-cheat multiplayer speed-math arena with server-authoritative scoring.

VORTEX 2K26 Global AI & VOYAGEHACK 4.0 Hackathons | *Team Leader — Team Eklavya* **2026**

- Directed cross-functional hackathon teams across global challenges, managing agentic AI workflows and rapid deployment.

FEATURED TECHNICAL PROJECTS

AI Smart Attendance System | *BCA 5th Sem Minor Project (Academic)* **2026**

Tech Stack: Python, FastAPI, OpenCV, React Native (Expo), SQLite, Docker Compose, Pytest, JWT, Haversine GPS

- **Geofenced Biometrics:** Architected dual-factor biometric verification combining Haversine GPS geofencing (strict 100m perimeter) with OpenCV facial feature matching to eliminate proxy punches.
- **Client & Admin:** Built complete React Native (Expo) client with offline punch caching and an administrative dashboard for real-time leave management and attendance CSV exports.
- **Testing & DevOps:** Engineered robust backend with 35 automated Pytest unit/integration tests; containerized application with Docker Compose for seamless reproducible deployment.

Enterprise Knowledge Graph RAG Copilot | *Academic Project* **2026**

Tech Stack: Python, FastAPI, Neo4j (Cypher), PostgreSQL (pgvector), Ollama, Next.js 14, Reciprocal Rank Fusion (RRF)

- **Tri-Hybrid Retrieval:** Designed an enterprise GraphRAG copilot combining dense vector embeddings with Neo4j multi-hop knowledge graph reasoning using Reciprocal Rank Fusion (RRF).
- **Anti-Hallucination:** Eliminated hallucinations through strict source citation provenance; integrated local Ollama containers enabling zero-cost private LLM inference.
- **Production Streaming:** Implemented real-time Server-Sent Events (SSE) streaming connected to a modern Next.js 14 App Router dashboard.

90-Day Workforce Attendance & Punctuality Analytics | *Academic Project* **2026**

Tech Stack: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Statistical Modeling

- **Statistical EDA:** Executed comprehensive exploratory data analysis (EDA) modeling 3,550 employee punch records over 90 days across 5 corporate departments.
- **Correlation Insights:** Quantified punctuality disparities (Engineering 82.4% vs. Sales 68.1%) and proved strong correlation (r = +0.67) between commute distance and tardiness.
- **Actionable Policy:** Formulated data-backed HR policy recommending 15-minute flexible grace periods, projected to eliminate 74% of transit-induced friction.

Dharohar 4D — 3D Cultural Heritage Digital Archive | *HACKCULT '26 Hackathon (IIT GN)* **2026**

Tech Stack: Three.js, React, WebGL, Next.js 14, WebXR AR, Python, FastAPI

- **WebXR Mobile AR:** Constructed interactive 4D digital twin archive enabling WebXR mobile AR exploration of historic monuments with primary source citations.
- **Dynamic 3D Engine:** Engineered dynamic time-of-day lighting engine (Sunrise, Golden Hour, Night floodlights) and clickable 3D architectural annotation pins.

MATHSPRINT — Multiplayer Classroom Speed-Math Arena | *SOU Mini Hackathon* **2025**

Tech Stack: React Native, Expo, TypeScript, Node.js, Firebase Realtime Database

- **Anti-Cheat Architecture:** Engineered high-intensity classroom speed-math battle platform with zero-trust server-authoritative anti-cheat scoring and synced clocks.
- **Live Telemetry:** Built live teacher telemetry command dashboard featuring real-time class rankings, streak multipliers, and granular topic diagnostics.

CERTIFICATIONS & PROFESSIONAL TRAINING

- Oracle Certified Associate: Agentic AI Foundations Associate (2026)
- Google Cloud: Introduction to Generative AI (2025)
- Infosys Springboard: Data Visualization with Tableau (2025)
- Akash Technolabs: Certified Android App Development Using Java (2024)
- Unlox: Generative AI Certification & UI/UX Design Internship (2024)